

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 06)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 A conservation organisation is converting 700 ha of arable farmland on the coast into a wetland reserve for breeding and wintering birds (as at Wallasea Island).

0 1 . 1 Explain why habitat area and shape are important in the design of the reserve. **[4 marks]**

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0 1 . 2 Explain how the reserve could be designed to provide a range of niches. **[3 marks]**

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0 1 . 3 Explain why some species may need to be deliberately introduced while others colonise naturally. **[2 marks]**

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0 1 . 4 Explain the value of a biological corridor linking the reserve to other wetlands. **[1 mark]**

10

0 2 . 1 Describe **three** methods of reducing seabird by-catch in long-line fisheries. **[3 marks]**

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0 2 . 2 Explain how restricting fishing effort and restricting fishing methods reduce overfishing. **[3 marks]**

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0 2 . 3 Describe how captive rearing and release ('population seeding') can increase wild populations of lobsters or oysters. **[2 marks]**

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0 2 . 4 Explain why collecting data on planktonic fish eggs and larvae helps to prevent population collapse. **[2 marks]**

10

03 . 1

Explain how overfishing of the giant triton mollusc damaged coral reefs in Australia.

[2 marks]

03 . 2

Describe **four** management practices used in marine protected areas to protect coral reefs.

[4 marks]

1

2

3

4

03 . 3

Explain why deep-water coral reefs recover much more slowly from trawling damage than tropical reefs.

[2 marks]

0	3	.	4
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Give **two** threats to deep-water reefs and **one** conservation measure.

[2 marks]

10

0	4	.	1
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Describe the main principles of the circular economy.

[6 marks]

0	4	.	2
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Explain what is meant by 'optimum production rather than maximum production', using a comparison between a natural ecosystem and an agroecosystem.

[2 marks]

04 . 3

Explain why international cooperation (eg through the EU or UN) is needed to implement circular economy initiatives.

[2 marks]

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05 . 1

Describe how agriculture alters the hydrological cycle.

[3 marks]

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05 . 2

Explain how each of the following consumer choices can reduce the environmental impact of food production.

[4 marks]

Seasonal / local food

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Organic food

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05 . 3

Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

[3 marks]

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10

0 6 . 1

Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

- A Close planting of a single non-indigenous species ☐
- B Mixed indigenous species with a mixed age structure ☐
- C Clear-felling on a 40-year rotation ☐
- D Removal of dead wood ☐

0 6 . 2

Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

1

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2

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5

07 . 1

Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

[6 marks]

Water availability

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Dissolved oxygen

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pH

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07 . 2

Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

[2 marks]

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07 . 3

Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

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08 . 1

Describe **four** ways in which the Wildlife and Countryside Act (1981) protects wildlife in the UK.

[4 marks]

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08 . 2

Explain the difference between CITES Appendix I and Appendix II, giving an example species for each.

[4 marks]

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08 . 3

Explain why controlling international trade can be more effective than trying to stop collection of a species in its own country.

[2 marks]

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09

Table 1 lists some feedback mechanisms that affect the Earth's climate.

09

1

Complete Table 1 by stating whether each mechanism is an example of **negative** or **positive** feedback.

[4 marks]

Table 1

Mechanism	Type of feedback
Higher temperatures increase cloud cover, raising albedo	
Melting permafrost releases methane	
Higher CO ₂ levels increase photosynthesis and carbon sequestration	
Melting ice reduces albedo	

09

2

Explain why natural systems regulated by negative feedback rarely reach tipping points.

[2 marks]

09

3

Using an example, explain how a large unusual event can cause a tipping point.

[2 marks]

09 . 4

Explain how the human response to negative feedback in agriculture can make food production less sustainable.

[2 marks]

10

10 . 1

Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

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2

 State what is meant by **biodiversity**.

Do not write outside the box

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

OR

1 1 . 2

Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

END OF QUESTIONS